



(12) **United States Patent**
Elie et al.

(10) **Patent No.:** **US 12,416,735 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **NAVIGATION METHOD AND SYSTEM**

(71) Applicants: **SAFRAN ELECTRONICS & DEFENSE**, Paris (FR); **M3 SYSTEMS**, L'Union (FR)

(72) Inventors: **Philippe Elie**, Paris (FR); **Vincent Bisgambiglia**, Paris (FR); **Elie Amani**, L'Union (FR); **Marc Pollina**, L'Union (FR)

(73) Assignee: **SAFRAN ELECTRONICS & DEFENSE**, Paris (FR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 511 days.

(21) Appl. No.: **17/792,956**

(22) PCT Filed: **Jan. 26, 2021**

(86) PCT No.: **PCT/EP2021/051693**

§ 371 (c)(1),

(2) Date: **Jul. 14, 2022**

(87) PCT Pub. No.: **WO2021/151864**

PCT Pub. Date: **Aug. 5, 2021**

(65) **Prior Publication Data**

US 2023/0043214 A1 Feb. 9, 2023

(30) **Foreign Application Priority Data**

Jan. 27, 2020 (FR) 2000771

(51) **Int. Cl.**

G01S 19/49 (2010.01)

G01S 5/02 (2010.01)

(Continued)

(52) **U.S. Cl.**

CPC **G01S 19/49** (2013.01); **G01S 5/0264** (2020.05); **G01S 19/46** (2013.01); **G01C 21/165** (2013.01); **G01S 19/45** (2013.01); **G01S 19/48** (2013.01)

(58) **Field of Classification Search**

CPC G01S 19/46; G01S 19/49; G01S 19/45; G01S 19/48; G01S 5/0045; G01S 5/0063; (Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,412,456 B2 * 4/2013 Roh G01S 19/49 701/472

8,860,609 B2 * 10/2014 Roh G01S 19/40 701/500

(Continued)

FOREIGN PATENT DOCUMENTS

CN 106646570 A * 5/2017 G01S 19/47

CN 109085626 A * 12/2018 G01S 19/425

OTHER PUBLICATIONS

Apr. 23, 2021 International Search Report issued in International Patent Application No. PCT/EP2021/051693.

(Continued)

Primary Examiner — Chuong P Nguyen

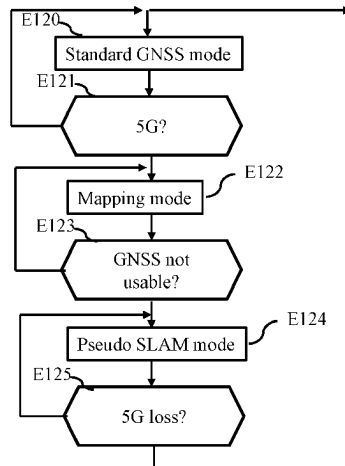
(74) *Attorney, Agent, or Firm* — Williams Mullen; R. Brian Drozd

(57)

ABSTRACT

Navigation device including an inertial navigation system coupled with a satellite navigation system, the information supplied by the satellite positioning system being used for adjusting the inertial navigation system, characterised in that the device further comprises means for measuring signals coming from base stations of a wireless cellular network, the satellite navigation system and the means for measuring signals coming from base stations of the wireless cellular network are coupled with each other by a tight coupling to

(Continued)



form a satellite navigator and/or base-station navigator implementing an estimator with simultaneous localisation and mapping, and in that the satellite navigation system and the means for measuring signals coming from base stations of the wireless cellular network are coupled to the inertial navigation system by a loose coupling for adjusting the inertial navigation system.

6 Claims, 8 Drawing Sheets

- (51) **Int. Cl.**
G01S 19/46 (2010.01)
G01C 21/16 (2006.01)
G01S 19/45 (2010.01)
G01S 19/48 (2010.01)
- (58) **Field of Classification Search**
 CPC ... G01S 5/0263; G01S 5/0264; G01C 21/165;
 G01C 21/02
 USPC 342/357.28, 357.29, 357.31, 357.32
 See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,746,329	B2 *	8/2017	Nichols	E02F 3/842
9,927,530	B2 *	3/2018	Boyarski	G01S 19/47
10,578,743	B1 *	3/2020	Kalathil	G01S 19/48
10,670,727	B2 *	6/2020	Revol	G01S 19/15
11,372,114	B2 *	6/2022	Carcanague	G01S 19/20
11,860,260	B2 *	1/2024	Carcanague	G01S 19/44
2016/0061614	A1 *	3/2016	Lee	G01C 21/28 701/446

OTHER PUBLICATIONS

Apr. 23, 2021 Written Opinion issued in International Patent Application No. PCT/EP2021/051693.

Javier Arribas et al. "A Technology-Agnostic GNSS/INS Real-Time Sensor Fusion Platform With Ultra Wide Band Cooperative Distance Measurements for Terrestrial Vehicle Navigation". GNSS 2018—Proceedings of the 31st International Technical Meeting of the Satellite Division of the Institute of Navigation (Ion GNSS+ 2018), The Institute of Navigation, 8551 Rixlew Lane Suite 360 Manassas, VA, 20109, USA, Sep. 28, 2018, pp. 1967-1984.

"Satellite Earth Stations and Systems (ses); Global Navigation Satellite System (GNSS) Based Location Systems; Minimum Performance and Features". Technical Report, European Telecommunications Standards Institute (ETSI), 650, Route des Lucioles; F-06921 Sophia-Antipolis; France, vol. SES SCN, No. V1.1, Sep. 1, 2012.

* cited by examiner

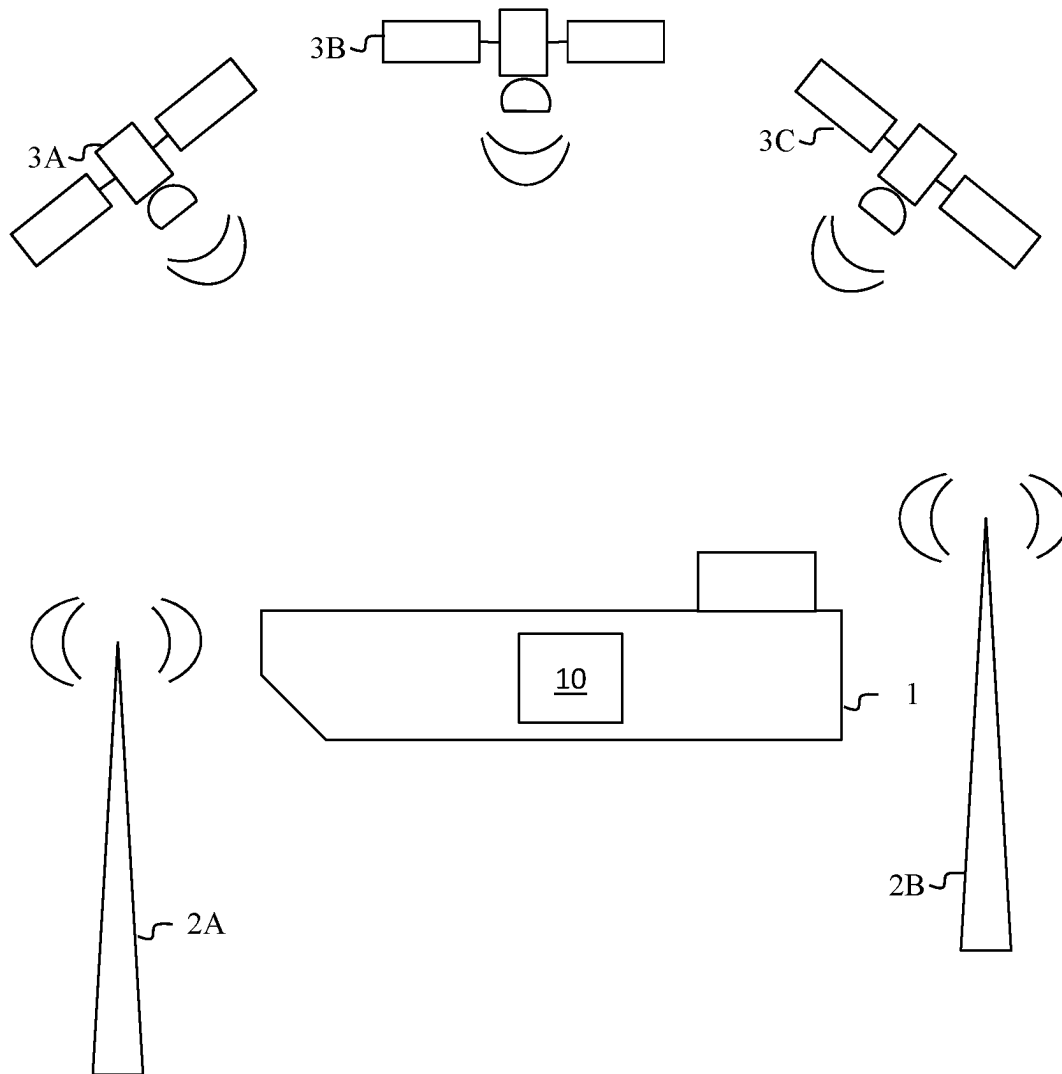


Fig. 1

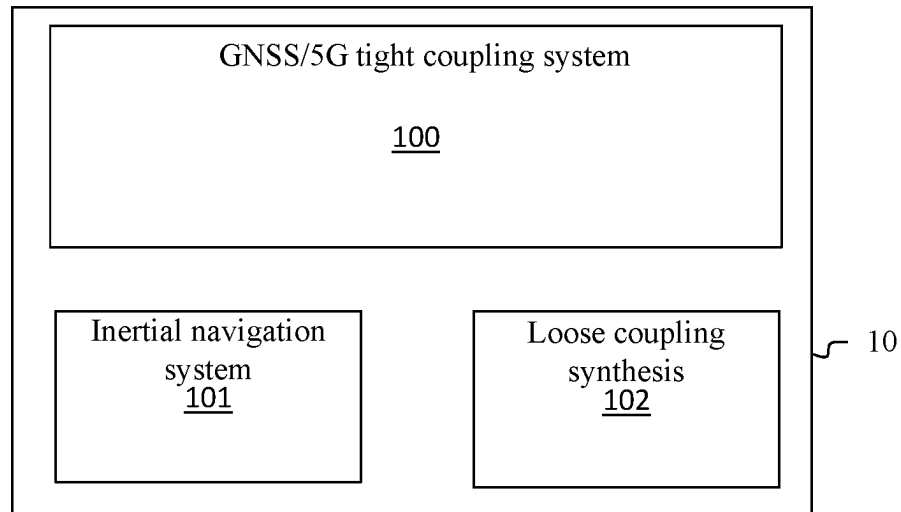


Fig. 2

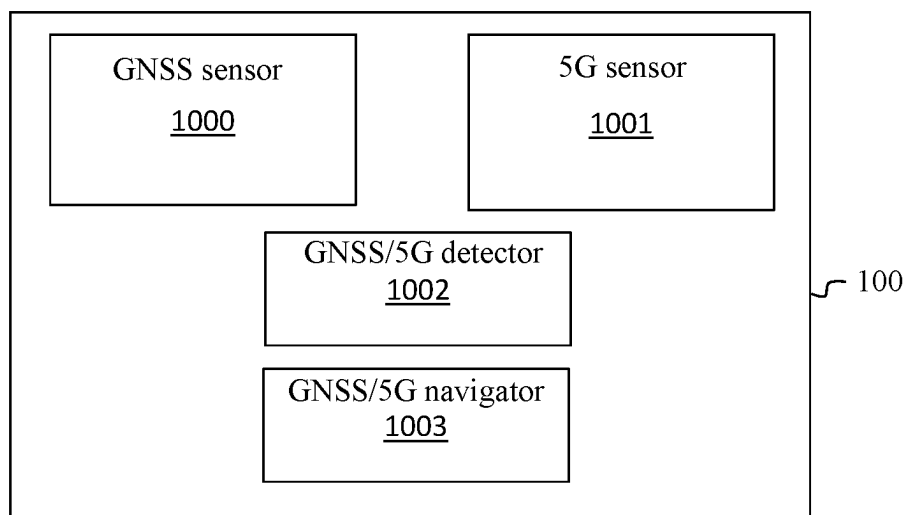


Fig. 3

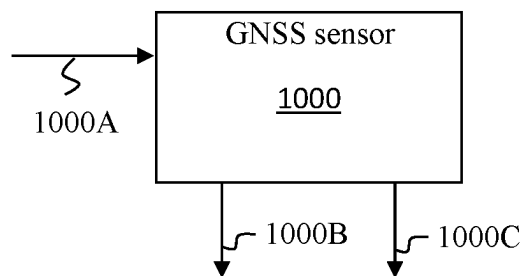


Fig. 4

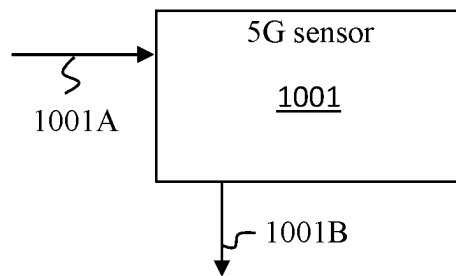


Fig. 5

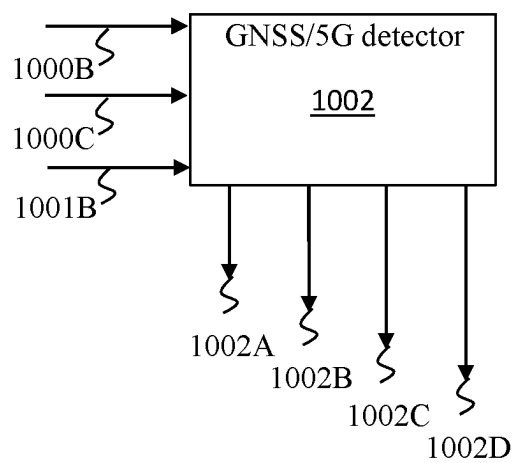


Fig. 6

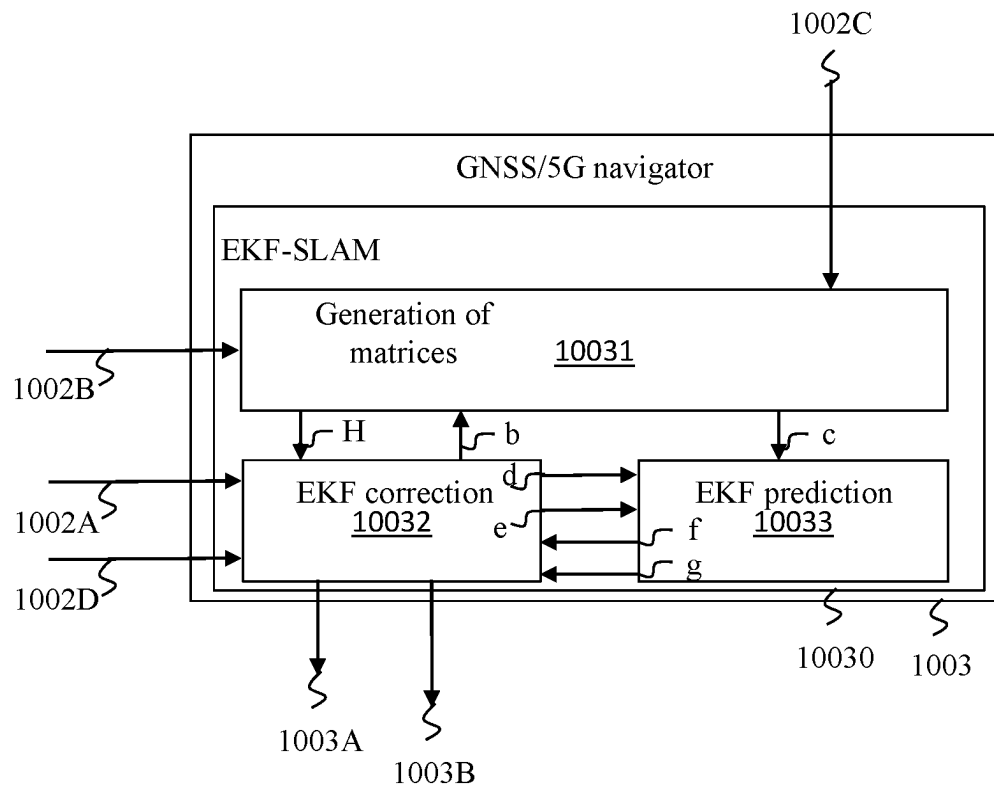


Fig. 7

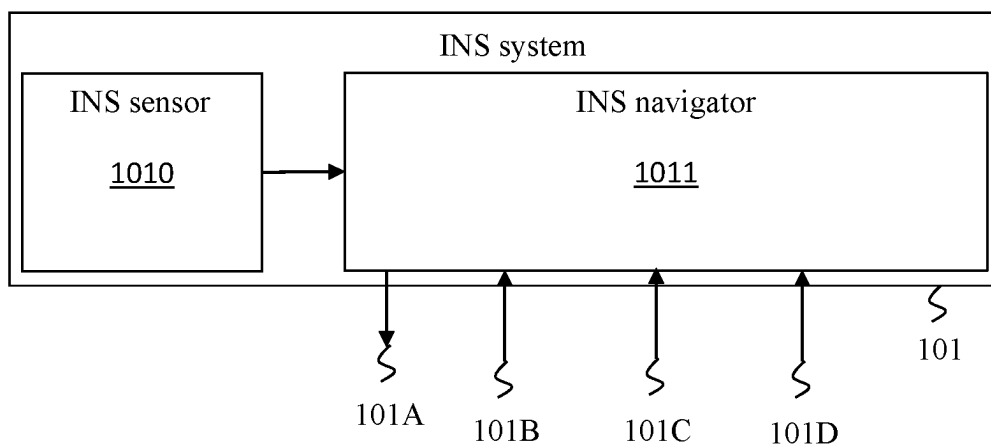


Fig. 8

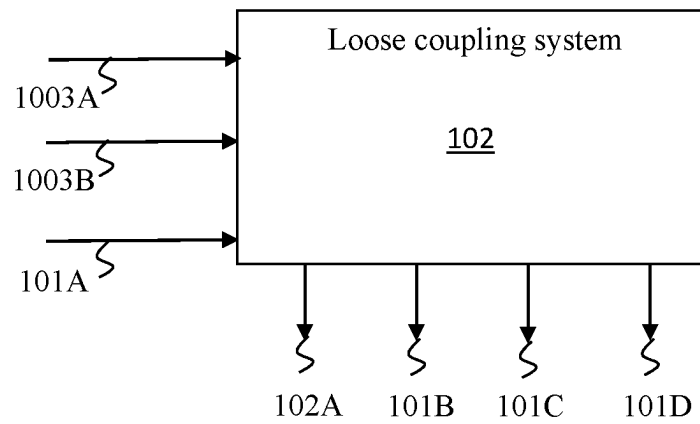


Fig. 9

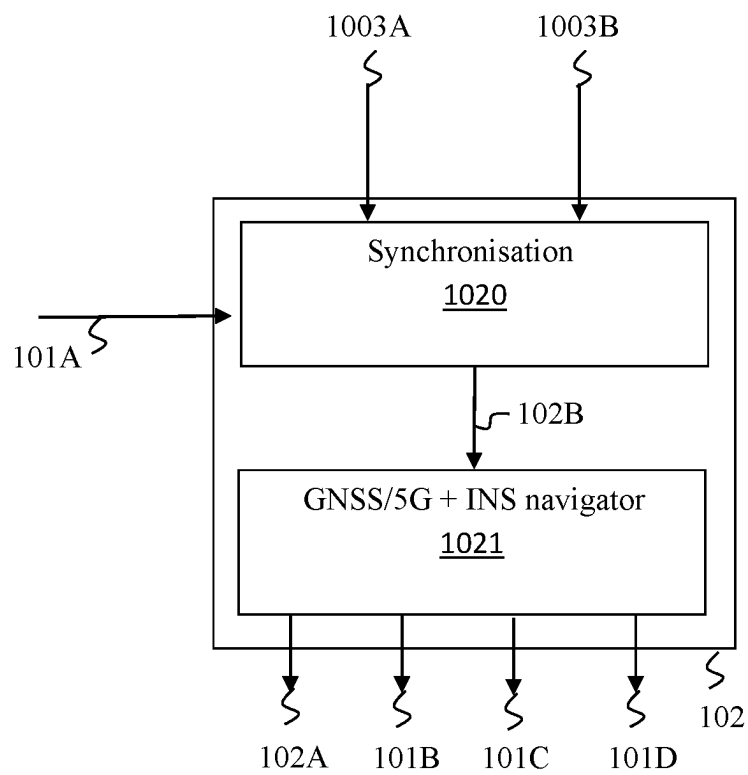


Fig. 10

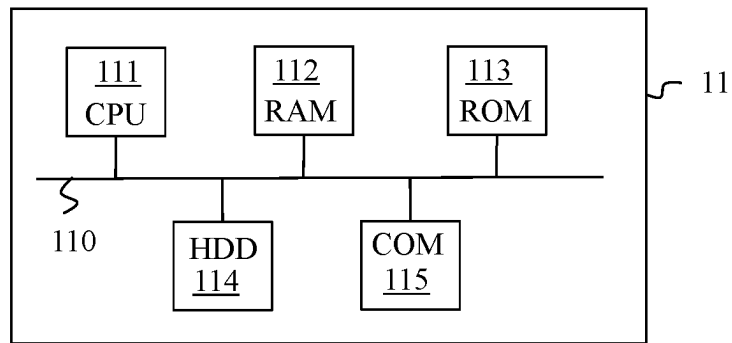


Fig. 11

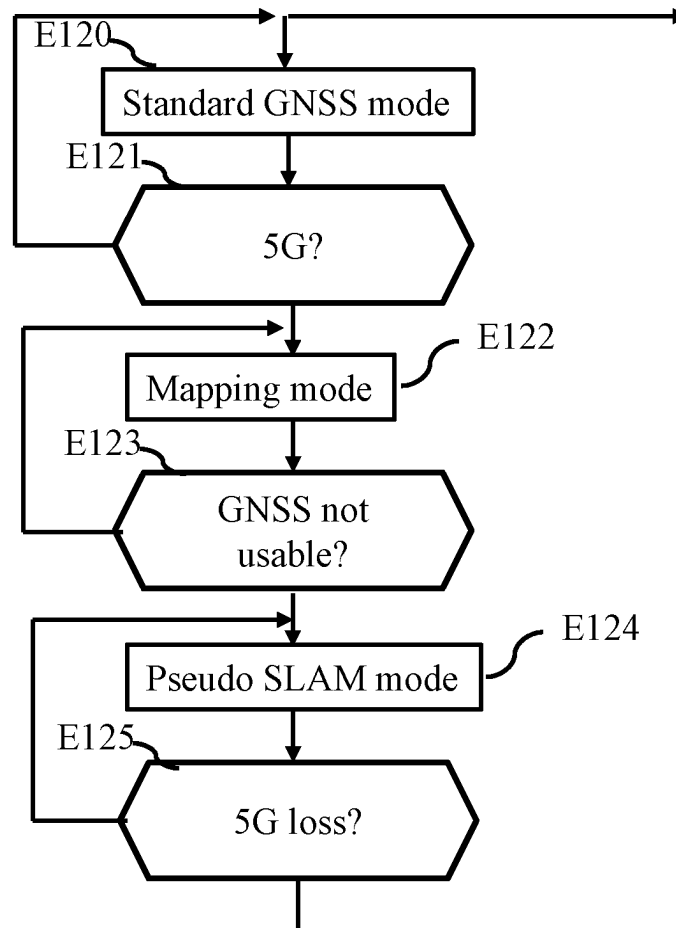


Fig. 12

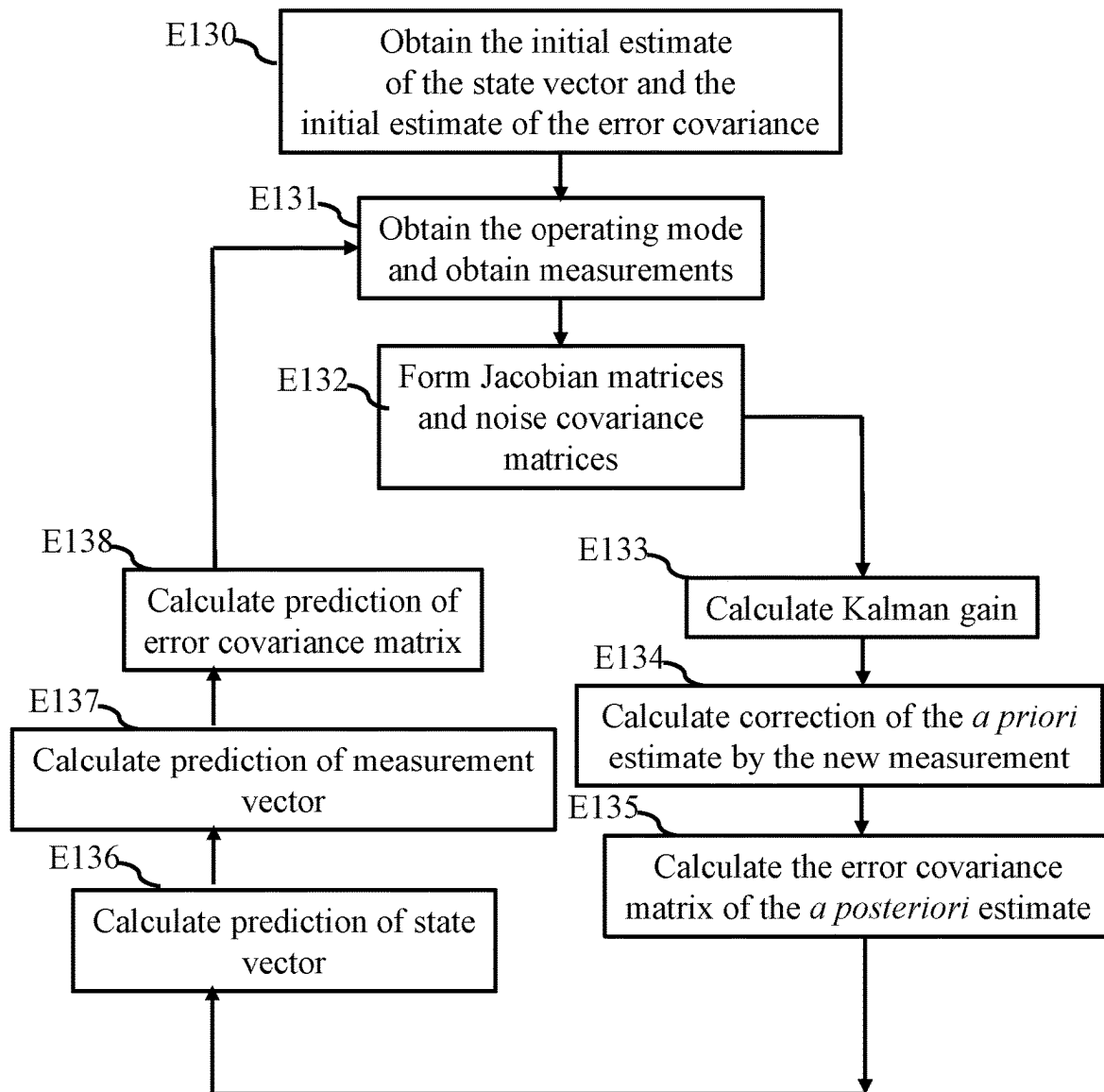


Fig. 13

NAVIGATION METHOD AND SYSTEM

TECHNICAL FIELD

The invention relates to a navigation method and system. 5

PRIOR ART

The use of inertial navigation systems, referred to as INS systems, is at the present time conventional, in particular in the field of air and sea navigation. INS systems are thus habitually installed in vehicles, such as aircraft or ships. INS systems use various sensors such as accelerometers or gyrometers. These sensors provide measurements (measurements of rotation, acceleration, etc.) an integration of which makes it possible to determine an attitude, a speed and a position of the vehicle during movement. 10

These sensors are however imperfect and have intrinsic errors or measurement biases liable to vary during a movement. These sensors are moreover subject to measurement noise. 15

One consequence of these imperfections is that INS systems prove to be precise over a short period but are, in the longer term, subject to significant drifts.

To overcome these drifts, some INS systems are coupled with a satellite positioning system of the GNSS type (Global Navigation Satellite System), referred to as GNSS system hereinafter, such as a GPS system (Global Positioning System), GLONASS (the Russian satellite navigation system), Galileo (the future European satellite navigation system) or Beidou (the future Chinese satellite navigation system). A so-called GNSS/INS hybrid system is then obtained, as described for example in the document EP2245479. 20

In a GNSS/INS hybrid system, the information supplied by the GNSS system is generally used to adjust the INS system. Conventionally, this adjustment is performed by means of a Kalman filter that optimises the navigation performances by estimating the respective errors of the GNSS system and of the INS system. 25

Two types of coupling are then distinguished: loose coupling and tight coupling.

In loose coupling, in addition to receiving navigation solutions (positions, speeds) of the INS system, the Kalman filter receives as an input navigation solutions (positions, speeds) supplied by the GNSS system.

In tight coupling, the GNSS system supplies as an input of the Kalman filter raw data such as pseudo-distances or pseudo-speeds in place of the navigation solutions.

At the present time, a majority of the GNSS/INS hybrid navigation systems use loose coupling because of its simplicity of implementation. 30

However, to be able to provide a navigation solution, a GNSS system must capture signals from at least a predetermined number of satellites, referred to as GNSS satellites, in general equal to four. In some environments, such as urban environments, the reception of the satellite signals is disturbed, which may at least temporarily prevent the GNSS system for capturing the predetermined number of GNSS satellites necessary for determining a navigation solution. These disturbances are in particular related to maskings of the GNSS satellites, multipaths and interference. When it encounters such disturbances, a GNSS system may be incapable of providing a navigation solution at the input of the Kalman filter. Without assistance from the GNSS system, a GNSS/INS hybrid system then becomes a simple INS system. 35

One advantage of a GNSS/INS hybrid system with tight coupling is that, even in the case of a number of GNSS satellites accessible to the GNSS system below the predetermined number of GNSS satellites, the GNSS/INS hybrid system can profit from the provision of the raw data supplied by the GNSS system for determining an improved navigation solution compared with an INS system alone. However, the management of the raw GNSS data, e.g. the calculation of correction of the clocks of the GNSS satellites, of the positions and of the speeds from ephemeris data, makes the GNSS/INS tight-coupling hybrid systems substantially more complex than the loose-coupling GNSS/INS hybrid systems. It is desirable to overcome these drawbacks of the prior art. It is in particular desirable to propose a method and a system that make it possible to overcome the defects of a loose-coupling GNSS/INS hybrid navigation system while keeping a complexity less than a tight-coupling GNSS/INS hybrid navigation system. 40

DISCLOSURE OF THE INVENTION

According to a first aspect, the present invention relates to a navigation device comprising an inertial navigation system coupled with a satellite navigation system, the information supplied by the satellite positioning system being used for adjusting the inertial navigation system, characterised in that the device further comprises means for measuring signals coming from base stations of a wireless cellular network, the satellite navigation system and the means for measuring signals coming from base stations of the wireless cellular network are coupled with each other by a tight coupling to form a satellite navigator and/or base-station navigator implementing an estimator with simultaneous localisation and mapping, and in that the satellite navigation system and the means for measuring signals coming from base stations of the wireless cellular network are coupled to the inertial navigation system by a loose coupling for adjusting the inertial navigation system. 45

The invention also relates to a navigation method comprising an inertial navigation system coupled with a satellite navigation system, the information supplied by the satellite positioning system being used for adjusting the inertial navigation system, characterised in that the method comprises the steps of:

measuring signals coming from base stations of a wireless cellular network,

coupling by tight coupling of the satellite navigation system and means for measuring signals coming from base stations of the wireless cellular network to form a satellite navigator and/or base-station navigator implementing an estimator with simultaneous localisation and mapping,

coupling by loose coupling of the satellite navigation system of the means for measuring signals coming from the base station of the wireless cellular network to the inertial navigation system to adjust the inertial navigation system. 50

Thus the present invention makes it possible to potentially have the largest amount of information available (GNSS and/or 5G) within a hybridisation architecture based on a loose coupling of the pseudo-distance information adjusting the inertial navigation and relying on a GNSS/5G tight coupling structure that makes it possible to optimise to the maximum possible extent the merging of the raw information coming from the two satellite sources via GNSS and 5G receivers. 55

3

According to a particular embodiment of the invention, the satellite and/or base station navigator operates in three operating modes, the first mode, referred to as mapping mode, being a mode wherein the pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network and signals from the satellites are available, the second mode, referred to as 5G mode, being a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming solely from base stations of the wireless cellular network are available, and the third mode, referred to as standard, being a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals solely of the signals of the satellites are available.

Thus a maximum quality of a continuous navigation function is obtained when only the GNSS is available or when only the cellular network is available or when the GNSS and the cellular network are available.

According to a particular embodiment of the invention, the device further comprises a matrix-generating module that forms a plurality of matrices comprising a transition matrix, a Jacobian matrix, a measurement-noise covariance matrix and a process-noise covariance matrix.

Thus, according to a particular embodiment of the invention, for each operating mode, the matrix-generating module forms a transition matrix, a Jacobian matrix, a measurement-noise covariance matrix and a process-noise covariance matrix that are different from the matrices formed for the other modes.

According to a particular embodiment of the invention, the navigation device comprises a Kalman filter.

According to a particular embodiment of the invention, the method comprises the step of determining an operating mode of the satellite and/or base-station navigator among three operating modes, and the first mode is a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network and signals from the satellites are available, the second mode is a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network are available and the third mode is a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals of the signals of the satellites are available.

According to a particular embodiment of the invention, the method furthermore comprises a matrix-generating step that forms a plurality of matrices comprising a transition matrix, a Jacobian matrix, a measurement-noise covariance matrix and a process-noise covariance matrix.

BRIEF DESCRIPTION OF THE DRAWINGS

The features of the invention mentioned above, as well as others, will emerge more clearly from the reading of the following description of an example embodiment, said description being made in relation to the accompanying drawings, among which:

FIG. 1 illustrates a context wherein the invention is implemented;

FIG. 2 illustrates schematically a hybrid navigation system with loose coupling between an INS system and a GNSS/5G tight coupling system according to the invention;

FIG. 3 illustrates schematically a GNSS/5G tight coupling system;

FIG. 4 illustrates schematically a GNSS system;

FIG. 5 illustrates schematically a 5G system;

FIG. 6 illustrates schematically a GNSS/5G detector;

4

FIG. 7 illustrates schematically a GNSS/5G navigator;

FIG. 8 illustrates schematically an INS navigation system;

FIG. 9 illustrates schematically a loose coupling system;

FIG. 10 illustrates schematically a loose coupling system between an INS system and a GNSS/5G tight coupling system;

FIG. 11 illustrates schematically an example of a hardware architecture of a processing module;

FIG. 12 illustrates schematically a method implemented by the processing module for selecting the operating mode of a navigator;

FIG. 13 illustrates schematically a method implemented by a navigator of the loose coupling system between an INS system and a GNSS/5G tight coupling system.

DETAILED DISCLOSURE OF EMBODIMENTS

The invention is described hereinafter in a maritime context wherein a navigation system implementing a method according to the invention is transported by a vehicle or carrier of the ship type. The invention could just as well apply in a different context such as an air or road context, the vehicle transporting the navigation system according to the invention being able to be an aircraft, a drone, a car or a lorry.

FIG. 1 illustrates a context wherein the invention is implemented. In FIG. 1, a ship 1 is moving in an environment comprising satellites (here satellites 3A, 3B and 3C are shown) of a constellation of GNSS satellites and base stations of a fifth generation cellular wireless communication system, referred to as 5G base stations (here two 5G base stations are shown). The ship 1 comprises a hybrid navigation system 10 with loose coupling between an INS system and a tight coupling GNSS/5G hybrid system, simply referred to as a GNSS/5G+INS hybrid navigation system hereinafter.

In one embodiment, the base stations could just as well be the base stations of a fourth generation telephony system (referred to as 4G base stations, or a set consisting of 5G base stations and 4G base stations).

FIG. 2 illustrates schematically the GNSS/5G+INS hybrid navigation system 10.

The GNSS/5G+INS hybrid navigation system 10 of FIG. 2 comprises a GNSS/5G tight coupling system 100, an INS system 101 and a loose coupling system 102. The various couplings are described hereinafter.

In one embodiment, the loose coupling system 102 receives corrected measurements from the INS system 101 and navigation solutions from the GNSS/5G tight coupling system 100.

In another embodiment, the loose coupling system 102 receives measurements from the INS system 101 and navigation solutions from the GNSS/5G tight coupling system 100 and, from this information, generates residual errors that it takes as an input measurement. FIG. 3 illustrates schematically the GNSS/5G tight coupling system 100.

The GNSS/5G tight coupling system 100 comprises a GNSS sensor 1000, a 5G sensor 1001, a GNSS/5G detector 1002 and a GNSS/5G navigator 1003.

The GNSS/5G detector 1002 defines, according to the output data that it receives, whether the GNSS/5G tight coupling system 100 can use only GNSS data, only 5G data or 5G data and GNSS data. The GNSS/5G detector 102 thus defines the mapping or 5G or standard operating mode.

The GNSS/5G navigator 1003 is based on a SLAM Kalman filter which, according to the operating mode, configures the SLAM Kalman filter. Thus the GNSS/5G

navigator **1003** defines, for each of the modes, the state and observation vectors, the matrices of the Kalman filter and more particularly their dimension.

A major advantage of the GNSS/5G tight coupling system **100** is that it uses a tight coupling between the GNSS sensor **1000** and the 5G sensor **1001**, thus allowing a deployment in a navigator (i.e. the GNSS/5G navigator **1003**) common to the GNSS sensor **1000** and to the 5G sensor **1001**, of a SLAM (Simultaneous Localisation and Mapping) estimator. The SLAM estimator is capable of simultaneously estimating states (i.e. positions, clock states) of the 5G base stations, and navigation solutions (i.e. positions, speeds and clock state) of the GNSS/5G tight coupling system **100**. Thus, as long as the GNSS/5G tight coupling system **100** captures at least a predetermined number of GNSS satellites N_{sat} (in general $N_{sat}=4$), it is capable of estimating the states (positions and clock states) of the 5G base stations. The GNSS/5G tight coupling system **100** therefore does not require external additional means for determining the positions and clock states of the 5G base stations. Once the states of the 5G base stations are known, when the number of satellites captured by the GNSS/5G tight coupling system **100** is less than N_{sat} , the GNSS/5G tight coupling system is capable of providing a navigation solution by compensating for the absence of GNSS signals by measurements coming from the 5G sensor **1001**.

The navigation solutions determined by the GNSS/5G tight coupling system **100** are next combined with measurements from the INS system **101** in the loose coupling system **102** to produce an improved navigation system and to correct any errors of the INS system **101**. FIG. 4 illustrates schematically the GNSS sensor **1000**. The GNSS sensor **1000** receives radio-frequency (RF) signals **1000A** from GNSS satellites. For each RF signal received from a GNSS satellite with sufficient quality, the GNSS sensor **1000** generates output data **1000B**, referred to as GNSS output data, comprising pseudo-distances and pseudo-speeds, referred to as GNSS pseudo-distances and pseudo-speeds, and a carrier-to-noise ratio C/N_0 , referred to as GNSS C/N_0 ratio, corresponding to said RF signal and estimations **1000C** of the position of said GNSS satellite.

FIG. 5 illustrates schematically the 5G sensor **1001**. The 5G sensor **1001** receives RF signals, referred to as 5G signals **1001A** from 5G base stations. For each RF signal received from a 5G base station with sufficient quality, the 5G sensor **1001** generates output data, referred to as 5G output data, comprising estimations **1001B** of pseudo-distances and pseudo-speeds, referred to as 5G pseudo-distances and pseudo-speeds, and of a carrier-to-noise ratio C/N_0 , referred to as 5G C/N_0 ratio, corresponding to said RF signal.

FIG. 6 illustrates schematically the GNSS/5G detector **1002**.

The GNSS/5G detector **1002** receives the GNSS and 5G output data comprising:

- the estimations **1000B** of the GNSS pseudo-distances and pseudo-speeds and of the GNSS ratios C/N_0 ;
- the estimations **1000C** of the positions of the GNSS satellites; and
- the estimations **1001B** of the 5G pseudo-distances and pseudo-speeds and of the 5G C/N_0 ratios.

The GNSS/5G detector **1002** then fulfils two roles. Its first role is to synchronise the GNSS output data with the 5G output data when the two types of data are available. For example, if several 5G output data are collected between two collections of GNSS output data, the 5G output data are averaged to reflect the average 5G output data over an

interval of time corresponding to the two collections of GNSS output data. The second role of the GNSS/5G detector **1002** is to define, according to the output data that it receives, whether the GNSS/5G tight coupling system **100** can use only GNSS data, only 5G data, or 5G data and GNSS data. The type of data used makes it possible to define an operating mode of the GNSS/5G+INS hybrid system **10** among a plurality of possible operating modes. The plurality of operating modes comprises a so-called mapping mode, a so-called 5G mode and a so-called standard (GNSS) mode. In the mapping mode, GNSS and 5G pseudo-distances, pseudo-speeds and C/N_0 ratios are available in a sufficient number.

In 5G mode, only the 5G pseudo-distances, pseudo-speeds and C/N_0 ratios are available.

In the standard mode, only the GNSS pseudo-distances, pseudo-speeds and C/N_0 ratios are available.

The GNSS/5G detector **1002** then supplies as an output measurements **1002A** of GNSS and/or 5G pseudo-speeds and pseudo-distances of the GNSS/5G tight coupling system **100**, estimations **1002B** of the GNSS and/or 5G C/N_0 ratios corresponding to the measurements **1002A**, positions of the GNSS satellites **1002C** corresponding to the measurements **1002A** and an identifier **1002D** indicating the data to be used by the GNSS/5G tight coupling system **100** (GNSS or 5G or GNSS and 5G).

FIG. 7 illustrates schematically the GNSS/5G navigator **1003**.

The GNSS/5G navigator **1003** uses the algorithm that will be described with reference to FIG. 12.

As mentioned above in relation to FIG. 2, the GNSS/5G navigator **1003** comprises a SLAM estimator. It is normal to use an extended Kalman filter (EKF), referred to as a EKF filter hereinafter, to solve a SLAM problem. The GNSS/5G navigator **1003** therefore comprises a module, referred to as a EKF-SLAM module **10030** solving the SLAM problem of the invention.

The EKF-SLAM module **10030** comprises a matrix-generating module **10031**, a correction module, referred to as an EKF correction module, **10032** and a prediction module, referred to as an EKF prediction module, **10033**.

The matrix-generating module **10031** is responsible for forming and using the positions of the satellites, the position and speed of the GNSS/5G tight coupling system **100** and the positions of 5G base stations, a plurality of matrices used in the navigation method that we describe hereinafter. This plurality of matrices comprises:

- a transition matrix F ;
- a Jacobian matrix H ;
- a measurement-noise covariance matrix R ;
- a process-noise covariance matrix Q .

The matrix-generating module **10031** receives as an input the estimations **1002B** of the GNSS and/or 5G C/N_0 ratios, the positions of the GNSS satellites **1002C** and corrected estimations **b** of GNSS and/or 5G pseudo-speeds and pseudo-distances of the GNSS/5G tight coupling system **100** supplied by the EKF correction module **10032**.

From these data received as an input, the matrix-generating module **10031** supplies the Jacobian matrix H at the input of the EKF correction module **10032**, and the matrices F , H , R and Q (reference c in FIG. 7) at the input of the EKF prediction module **10033**.

The EKF correction module **10032** receives as an input the estimations **1002A** of GNSS and/or 5G pseudo-speeds and pseudo-distances of the GNSS/5G tight coupling system **100**, the identifier **1002D** indicating the data to be used by the GNSS/5G tight coupling system **100**, the Jacobian

matrix **H** and, coming from the EKF prediction module **10033**, a prediction **f** of an error state vector of the GNSS/5G tight coupling system **100** and a prediction **g** of an error covariance matrix associated with said state vector.

From these data, the EKF correction module **10032** supplies to the EKF prediction module **10033** an a posteriori estimate (i.e. after correction) **d** of the error state vector of the GNSS/5G tight coupling system and an a posteriori estimate **e** of the error covariance matrix associated with said state vector. Moreover, the EKF correction module **10032** generates an identifier **1003A** indicating an operating mode of the GNSS/5G tight coupling system **100** and a navigation solution (referred to as GNSS/5G navigation solution, **1003B**).

The EKF prediction module **10033** receives as an input the matrices **F**, **H**, **R** and **Q** (reference **c**), the a posteriori estimate **d** of the error state vector of the GNSS/5G tight coupling system and the a posteriori estimate **e** of the error covariance matrix associated with said state vector.

From these data, the EKF prediction module **10033** supplies to the EKF correction module **10032** the prediction **f** of an error state vector of the GNSS/5G tight coupling system **100** and the prediction **g** of an error covariance matrix associated with said state vector.

FIG. 8 illustrates schematically the INS navigation system **101**.

The INS navigation system **101** uses the algorithm that will be described with reference to FIG. 13.

The INS system **101** comprises an INS sensor **1010** and an INS navigator **1011**. The INS sensor **1010** comprises for example at least one accelerometer and at least one gyrometer each providing measurements to the INS navigator **1011**.

Like a conventional INS navigator, the INS navigator **1011** receives the measurements from the INS sensor **1010** and generates a navigation solution, referred to as INS navigation solution, **101A**. However, the INS navigator **1011** also receives, from the loose coupling system **102**, position and speed corrections **101D**, attitude corrections **101C** and corrections (bias, scale factor of the accelerometer and of the gyrometer) **101B** to be applied to the measurements supplied by the INS sensor **1010**.

FIG. 9 illustrates schematically the loose coupling system **102**.

The loose coupling system **102** receives the identifier **1003A** indicating the operating mode of the GNSS/5G tight coupling system **100** and the GNSS/5G hybrid navigation solution **1003B** supplied by said GNSS/5G tight coupling system **100**. The loose coupling system also receives the INS navigation solution **101A** supplied by the INS system **101**.

From these data, the loose coupling system **102** supplies the corrections **101B**, **101C** and **101D** to the INS system **101** and generates a navigation solution, referred to as GNSS/5G+INS hybrid navigation solution **102A**. The GNSS/5G+INS hybrid navigation solution is the navigation solution supplied by the GNSS/5G+INS hybrid navigation system **10**.

FIG. 10 illustrates schematically a detail of the loose coupling system **102**.

The loose coupling system **102** comprises a synchronisation module **1020** and a navigator, referred to as a GNSS/5G+INS navigator, **1021**. The synchronisation module **1020** is responsible for supplying to the GNSS/5G+INS navigator **1021** navigation solutions **102B** that it obtains from the navigation solutions **101A** and **1003B** coming respectively from the INS system **101** and from the GNSS/5G navigator **1003**. However, the INS system **101** and the GNSS/5G tight

coupling system **100** do not generally generate navigation solutions with the same frequency. The synchronisation module **1020** is then responsible for synchronising the INS navigation solutions with the GNSS/5G navigation solutions. For example, when the INS navigation solutions are supplied with a higher frequency than the GNSS/5G navigation solutions, the synchronisation module **1020** calculates GNSS/5G solutions that are intermediate between two GNSS/5G solutions supplied by the GNSS/5G tight coupling system **100**, each intermediate GNSS/5G solution being aligned timewise on an INS solution supplied by the INS system **101**. From these synchronised INS and GNSS/5G navigation solutions, the GNSS/5G+INS navigator **1021** generates corrections **101B**, **101C** and **101D** and the GNSS/5G+INS hybrid navigation solutions **102A**.

FIG. 11 illustrates schematically an example of a hardware architecture of a processing module **11**.

The modules included in the GNSS/5G+INS hybrid navigation system **10**, including the GNSS/5G detector **1002**, the matrix-generating module **10031**, the EKF correction module **10032**, the EKF prediction module **10033**, the synchronisation module **1020**, the GNSS/5G+INS navigator **1021**, each comprise a processing module **11**.

According to the example of hardware architecture shown in FIG. 11, the processing module **11** then comprises, connected by a communication bus **110**: a processor or CPU (central processing unit) **111**; a random access memory RAM **112**; a read only memory ROM **113**; a storage unit such as a hard disk or a storage medium reader, such as an SD (Secure Digital) card reader **114**; at least one communication interface **115** enabling the processing module **11** to communicate with other modules in the GNSS/5G+INS hybrid navigation system **10**.

The processor **111** is capable of executing instructions loaded in the RAM **112** from the ROM **113**, from an external memory (not shown), from a storage medium (such as an SD card), or from a communication network. When the GNSS/5G+INS hybrid navigation system **10** is powered up, the processor **111** is capable of reading instructions from the RAM **112** and executing them. These instructions form a computer program causing the complete or partial execution, by the processor **111**, of the methods executed by the GNSS/5G+INS hybrid navigation system modules **10** comprising said processing module **11**.

These methods can be implemented in software form by executing a set of instructions by a programmable machine, for example a DSP (digital signal processor) or a microcontroller, or be implemented in hardware form by a machine or a dedicated component, for example an FPGA (field-programmable gate array) or an ASIC (application-specific integrated circuit).

For example, when the present invention is implemented in a ship in a port-approach phase or in a coming-alongside phase, the measurements determine the operating mode of the GNSS/5G+INS hybrid receiver **10**. For example, when the GNSS measurements are partially masked and/or suffer partially from multipaths, the standard mode is selected.

When the GNSS measurements are completely or very strongly masked and/or completely suffer very greatly from multipaths, the 5G mode is selected.

When the 5G measurements are completely very greatly masked and/or completely suffer very greatly from multipaths, the INS/GNSS mode is selected.

In mapping mode, the GNSS and 5G pseudo-distances are available. The state vector of the EKF-SLAM estimator is made from the position, the speed, the bias and the drift of

the clock of the receiver of the hybrid receiver, as well as positions, biases and clock drifts of the 5G stations.

Between the corrections, the EKF-SLAM estimator produces the state prediction $\hat{X}_k^-$ and the prediction error covariance P_k^- by means of the transition models, as well as clock models of the receiver. When pseudo-distance and pseudo-speed measurements are available, the EKF-SLAM estimator produces an update $\hat{X}_k$ of the state estimate and an update P_k of the associated estimation error covariance.

In 5G mode, only the 5G pseudo-distances are available. The state vector of the estimator is made from the position and from the speed of the receiver as well as from the difference between the biases and clock drifts of the receiver on the one hand and of each 5G station on the other hand.

The 5G mode considered here is a restricted mode in that the positions of the 5G base stations as well as the bias and the clock drift of the receiver estimated in the mapping mode are used. As soon as the GNSS measurements become available again, the bias and the clock drift as well as the positions of the 5G stations are reintroduced into the state vector and mapping mode is gone into again, thus making it possible to refine the positions of the 5G base stations.

Between the corrections, the EKF-SLAM estimator produces a state prediction $\hat{X}_k^-$ and the prediction error covariance P_k^- by means of the transition models as well as clock models of the 5G base stations.

In GNSS mode, referred to as standard mode, only the GNSS pseudo-distances are available. The 5G measurements are unavailable following disturbances related either to the environment or to the absence of the 5G network.

The state vector of the EKF-SLAM estimator is made from the position, the speed, the bias and the drift of the clock of the receiver. Between the corrections, the EKF-SLAM estimator produces a state prediction $\hat{X}_k^-$ and the prediction error covariance P_k^- by means of the transition models as well as clock models of the receiver. When measurements of GNSS and/or 5G pseudo-distances and pseudo-speeds are available, the EKF-SLAM estimator produces an update $\hat{X}_k$ of the state estimate and an update P_k of the associated estimation error covariance.

Outside the modes, each 5G signal emanates from a spatially stationary base station and its state vector consists of its three-dimensional position coordinates $r_s=[x, y, z]^T$ and of its clock states $x_{hor,s}=[c\delta t_s, c\dot{\delta t}_s]$ where c is the celerity, δt_s the clock bias of the 5G base station and $\dot{\delta t}_s$ the clock drift of the 5G base station.

The dynamic range of the 5G signal is described by the following model: $(x_s)_{k+1}=F_s(x_s)_k+(w_s)_k$

where $x_s=r_s^T, x_{hor,s}^T|T, F_s=\text{diag}[I_{3 \times 3}, F_{hor}]$, w_s is the noise of the process modelled as white noise with zero mean and covariance

$$Q_s = \text{diag}[0_{3 \times 3}, c^2 Q_{hor,s}], F_{hor} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix},$$

$$Q_{hor,s} = \begin{bmatrix} S_{w\delta t_s} T + S_{w\dot{\delta t}_s} \frac{T^3}{3} & S_{w\dot{\delta t}_s} \frac{T^2}{2} \\ S_{w\dot{\delta t}_s} \frac{T^2}{2} & S_{w\delta t_s} T \end{bmatrix}$$

and T is the constant sampling interval.

The terms $S_{w\delta t_s}$ and $S_{w\dot{\delta t}_s}$ are the process noise spectra of the bias and of the clock drift of the 5G station respectively.

The terms $S_{w\delta t_s}$ and $S_{w\dot{\delta t}_s}$ can be associated with the power law coefficients, $\{h_\alpha\}_{\alpha=-2}^2$, which were demonstrated during laboratory experiments as characterising the power spectral density of the fractional frequency-difference values $y(t)$ of an oscillator with respect to the nominal frequency according to $S_y(f)=\sum_{\alpha=-2}^2 h_\alpha f^\alpha$. A usual approximation uses only the parameters h_0 and h_{-2} , so that

$$S_{w\delta t_s} \approx \frac{h_{0s}}{2} \text{ and } S_{w\dot{\delta t}_s} \approx 2\pi^2 h_{-2s}.$$

A state vector x_r of the GNSS/5G tight coupling system **100** consists of three-dimensional position coordinates $r_r=[X,Y,Z]^T$ of said system **100**, a three-dimensional speed of said system **100** $\dot{r}_r=[\dot{X},\dot{Y},\dot{Z}]^T$ and clock states of the receiver $x_{hor,r}=[c\delta t_r, c\dot{\delta t}_r]^T$: $x_r=[r_r^T, \dot{r}_r^T, x_{hor,r}^T]^T$

The states of the receiver change over time in the following manner:

$$(x_r)_{k+1} = F_r(x_r)_k + (w_r)_k$$

$$\text{where } F_r = \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & 0_{3 \times 2} \\ 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix},$$

T is the constant sampling interval of the measurements, w_r is the process noise vector model as modelled as a white noise with zero mean and covariance $Q_r=\text{diag}[Q_p, c^2 Q_{hor,r}]$. diag denotes the diagonal concatenation of blocks of matrices.

$$Q_r = \begin{bmatrix} S_p \frac{T^3}{3} & 0 & 0 & S_p \frac{T^2}{2} & 0 & 0 & 0 & 0 \\ 0 & S_p T & 0 & 0 & S_p \frac{T^2}{2} & 0 & 0 & 0 \\ S_p \frac{T^2}{2} & 0 & S_p T & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p \frac{T^3}{3} & 0 & S_p \frac{T^2}{2} & 0 & 0 \\ 0 & S_p \frac{T^2}{2} & 0 & 0 & S_p \frac{T^3}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p \frac{T^2}{2} & 0 & S_p T & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & c^2 S_{w\delta t_r} T + c^2 S_{w\dot{\delta t}_r} \frac{T^3}{3} & c^2 S_{w\dot{\delta t}_r} \frac{T^2}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & c^2 S_{w\dot{\delta t}_r} \frac{T^2}{2} & c^2 S_{w\delta t_r} T \end{bmatrix},$$

-continued

$$Q_{hor,r} = \begin{bmatrix} S_{w\delta t_r} T + S_{w\delta t_r} \frac{T^3}{3} & S_{w\delta t_r} \frac{T^2}{2} \\ S_{w\delta t_r} \frac{T^2}{2} & S_{w\delta t_r} T \end{bmatrix},$$

$$Q_{pv} = \begin{bmatrix} S_p \frac{T^3}{3} & 0 & 0 & S_p \frac{T^2}{2} & 0 & 0 \\ 0 & S_p T & 0 & 0 & S_p \frac{T^2}{2} & 0 \\ S_p \frac{T^2}{2} & 0 & S_p T & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p \frac{T^3}{3} & 0 & S_p \frac{T^2}{2} \\ 0 & S_p \frac{T^2}{2} & 0 & 0 & S_p \frac{T^3}{3} & 0 \\ 0 & 0 & 0 & S_p \frac{T^2}{2} & 0 & S_p T \end{bmatrix}$$

S_p is the speed noise power spectrum, $S_{w\delta t_r}$ and $S_{w\delta t_s}$ are the power spectra of the process noises of the bias and of the clock drift of the receiver respectively with

$$S_{w\delta t_r} = \frac{h_{0r}}{2} \text{ and } S_{w\delta t_s} \approx 2\pi^2 h_{-2r}.$$

For example, for a clock using a quartz oscillator, $h_{0r} = 2^{-19}$ seconds, and $h_{-2r} = 2^{-20}$ Hz.

The observation of pseudo-distance made by the GNSS/5G receiver on the n^{th} 5G base station at the instant k is linked to the states of the receiver and of the 5G base station:

$$(z_{s_n})_k = (\|r_r - r_{s_n}\|)^2 + c \cdot [\delta t_r - \delta t_{s_n}]_k + (v_{s_n})_k$$

where $r_r = [X, Y, Z]^T$ is the position of the receiver, $r_{s_n} = [x, y, z]^T$ is the position of the n^{th} 5G gNodeB, v_{s_n} is modelled as a Gaussian white noise with zero mean and variance σ_{α}^2 .

The observation of pseudo-speed (delta-distance) made by the GNSS/5G receiver with respect to the n^{th} 5G base station at the instant k is linked to the states of the receiver and of the 5G base station by:

$$(z_{s_n})_k = \left(\frac{1 - \frac{1}{c} [V_{s_n} - \dot{r}_r \cdot \hat{a}_{s_n}^T]}{1 + (\delta t_r - \delta t_{s_n})} \lambda \right) + (w_{s_n})_k \text{ where } \hat{a}_{s_n}^T \cong \frac{r_r - r_{s_n}}{\|r_r - r_{s_n}\|}$$

is the LOS vector of the 5G base station at the receiver, $\dot{r}_r = [\dot{X}, \dot{Y}, \dot{Z}]^T$ is the predicted speed of the receiver, $V_{s_n} = 0$ is the zero speed of the 5G transmitter as it is stationary in this framework, c is the celerity, δt_r the clock drift of the receiver and δt_{s_n} the clock drift of the 5G station, λ is the wavelength of the nominal carrier frequency of the 5G base station, w_{s_n} is a Gaussian white noise with zero mean and variance σ_{β}^2 .

The observation of pseudo-distance made by the GNSS/5G receiver on the m^{th} GNSS satellite after correction of the ionospheric and tropospheric delays is given by:

$$(z_{sv_m})_k = (\|r_r - r_{sv_m}\|)^2 + c \cdot [\delta t_r - \delta t_{sv_m}]_k + (v_{sv_m})_k$$

where $z_{sv_m} = z_{sv_m}' - c\delta t_{iono} - c\delta t_{tropo}$, δt_{iono} and δt_{tropo} are the ionospheric and tropospheric delays respectively, z_{sv_m}' is the uncorrected pseudo-distance, and v_{sv_m} is modelled as a Gaussian white noise with zero mean and variance σ_{γ}^2 .

The observation of pseudo-speed (delta-distance) made by the GNSS/5G receiver with respect to the m^{th} GNSS satellite at the instant k is linked to the states of the receiver by:

$$(z_{sv_m})_k = \left(\frac{1 - \frac{1}{c} [V_{sv_m} - \dot{r}_r \cdot \hat{a}_{sv_m}^T]}{1 + (\delta t_r - \delta t_{sv_m})} \lambda_L \right) + (w_{sv_m})_k \text{ where } \hat{a}_{sv_m}^T \cong \frac{r_r - r_{sv_m}}{\|r_r - r_{sv_m}\|}$$

is the LOS vector of the satellite at the receiver, $\dot{r}_r = [\dot{X}, \dot{Y}, \dot{Z}]^T$ is the predicted speed of the receiver, V_{sv_m} is the speed of the satellite, c is the celerity, δt_r the clock drift of the receiver and δt_s the clock drift of the satellite, λ_L is the wavelength of the nominal carrier frequency of the satellite, w_{sv_m} is a Gaussian white noise with zero mean and variance σ_{δ}^2 .

The navigation algorithm uses an extended Kalman filter as estimator with an addition of complexity related to the fact that the states of the base stations $\{x_{s_n}\}_{n=1}^N$ are simultaneously estimated with the states of the GNSS/5G receiver.

In the mapping mode, the extended Kalman filter (EKF) produces an estimate $\hat{x}_k \cong E[x_k | \{Z_i\}_{i=1}^k]$ of x_k , and an error covariance estimate $P_k \cong E[\Delta x_k \Delta x_k^T]$ where $x_k \cong [x_r^T, x_{s_1}^T, \dots, x_{s_N}^T]^T$, $Z_k \cong [z_{sv}^T, z_s^T]^T$, $z_{sv} \cong [z_{sv_1}, \dots, z_{sv_M}]^T$, $z_s \cong [z_{s_1}, \dots, z_{s_N}]^T$, and k is the last instant at which measurements were taken. In the implementation of the EKF estimator, the position, speed, bias and clock drift errors of the receiver, and the position, speed, bias and clock drift errors of the 5G base stations are estimated. The states are subsequently updated at the instant k by adding the errors estimated at the instant k to the states estimated at the instant $k-1$, i.e.

$$\hat{x}_k = \hat{x}_{k-1} + \Delta x_k.$$

The state vector of the EKF estimator $\Delta x_k \cong x - \hat{x}_k$ at the instant k , x being the true vector of parameters to be estimated and $\hat{x}_k$ being the estimate of the parameter vector is given by

$$\Delta x_k = [\Delta r_r^T, \Delta \dot{r}_r^T, \Delta v_{hor,s}^T, \Delta r_{s1}^T, \Delta x_{hor,s1}^T, \dots, \Delta x_{hor,sN}^T]^T$$

where the position, speed and clock states are defined in the form of conventional additive error. To simplify the notation, we shall consider that the estimated error state vector is denoted $\hat{X}_k$ for the a posteriori estimation or $\hat{X}_k^-$ for the a priori estimation instead of Δx_k . The a posteriori estimate of the error state vector of the EKF estimator at the instant k is then denoted as follows: $\hat{X}_k = [\hat{r}_r^T, \hat{\dot{r}}_r^T, \hat{v}_{hor,s}^T, \hat{r}_{s1}^T, \hat{x}_{hor,s1}^T, \dots, \hat{x}_{hor,sN}^T]^T$

In the 5G mode, the measurements z_{sv} are not available. The estimates of the clock states of the receiver $\hat{x}_{hor,r}$ and the

13

position states of the 5G base stations $\hat{r}_{sn}$ are no longer calculated but the biases relating to 5G clocks with respect to the clock of the receiver $\Delta\hat{t}_n \equiv \hat{t}_r - \hat{t}_{sn}$, with $n=1$ to N , are estimated.

At the instant t_0 the states $\hat{x}_{hor,r}$, $\{\hat{r}_{sn}\}_{n=1}^N$ and $\{\hat{x}_{hor,sn}\}_{n=1}^N$ are removed from the estimator and the estimates of the relative clock states $\Delta\hat{x}_{horn}$ are initialised in the following manner: $\Delta\hat{x}_{horn} = [\Delta\hat{t}_n, \Delta\hat{t}_n]^T$, $n=1$ to N with $\Delta\hat{t}_n \equiv \hat{t}_r - \hat{t}_{sn}$ and $\Delta\hat{t}_n \equiv \Delta\hat{t}_r - \Delta\hat{t}_{sn}$.

The difference between the clock state vectors of the receiver and of the base station $\Delta\hat{x}_{hor} \equiv \hat{x}_{hor,r} - \hat{x}_{hor,s}$ change in accordance with the equation $(\Delta\hat{x}_{hor})_{k+1} \equiv (F_{hor})_k (\Delta\hat{x}_{hor})_k + (Fw_{hor})_k$ where $w_{hor} \equiv \hat{x}_{hor,r} - \hat{x}_{hor,s}$ is a white noise with zero mean and covariance $Q_{hor} \equiv Q_{hor,r} + Q_{hor,s}$.

The error $\Delta x'_k \equiv x'_k - \hat{x}_k$ in estimating the new state vector x' and the corresponding error covariance matrix are initialised in the following manner: $\Delta x'_k = D\Delta x_k$ and $P'_k = DP_k D^T$ where D is the matrix that transforms Δx_k to $\Delta x'_k = [\Delta\hat{r}_r^T, \Delta\hat{r}_r^T, \Delta\Delta x_{hor1}^T, \dots, \Delta\Delta x_{horN}^T]^T$.

The new error state vector is denoted $\hat{x}'_k = [\hat{r}_r^T, \hat{r}_r^T, \Delta\hat{x}_{hor1}^T, \dots, \Delta\hat{x}_{horN}^T]^T$ so that $\hat{x}'_k = D\hat{x}_k$. If the vector $\hat{x}_k$ is of dimension $U \times 1$ and the vector $\hat{x}'_k$ is of dimension $V \times 1$ then the D matrix is of dimension $V \times U$. D is calculated at each change of mode according to the number of 5G base stations available.

Between two correction instants, the estimator uses the dynamic-range models described previously for propagating the estimates of the states $\hat{x}_k$ or $\hat{x}'_k$, supplying the prediction of the corresponding error covariance matrix, and uses the observation model.

In mapping mode, the state prediction is made:

$$\hat{x}'_k = F_k \hat{x}_{k-1} \text{ with } \hat{x}'_k = \begin{bmatrix} \hat{r}_r^T, \hat{r}_r^T, \hat{x}_{hor,r}^T, \hat{r}_{s1}^T, \hat{x}_{hor,s1}^T, \dots, \hat{r}_{sN}^T, \hat{x}_{hor,sN}^T \end{bmatrix}^T$$

$$F_k = \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & 0_{3 \times 2} & 0_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 3} & 0_{3 \times 2} \\ 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 2} & 0_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} & 0_{2 \times 3} & 0_{2 \times 2} & \dots & 0_{2 \times 3} & 0_{2 \times 2} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 2} & I_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & 0_{2 \times 2} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} & \dots & 0_{2 \times 3} & 0_{2 \times 2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 2} & 0_{3 \times 3} & 0_{3 \times 2} & \dots & I_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & 0_{2 \times 2} & 0_{2 \times 3} & 0_{2 \times 2} & \dots & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix}$$

Note that

$$F_k \equiv \text{diag}[F_r, F_s, \dots, F_s], \text{ where } F_s = \text{diag}[I_{3 \times 3}, F_{hor}],$$

$$F_{hor} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, F_r = \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & 0_{3 \times 2} \\ 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix}.$$

Likewise, the error covariance prediction matrix is thus given by:

$$P_k^- = F_k P_{k-1} F_k^T + Q_k$$

$$F_k \equiv \text{diag}[F_r, F_s, \dots, F_s]$$

$$Q_k \equiv \text{diag}[Q_r, Q_s, \dots, Q_s]$$

14

In 5G mode, the following state prediction is made:

$$\hat{x}'_k = F'_k \hat{x}'_{k-1}$$

With $\hat{x}'_k = [\hat{r}_r^T, \hat{r}_r^T, \Delta\hat{x}_{hor1}^T, \dots, \Delta\hat{x}_{horN}^T]^T$

$$F'_k \equiv \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 2} \\ 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} & \dots & 0_{2 \times 3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_{2 \times 3} & 0_{2 \times 3} & 0_{2 \times 2} & \dots & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix}$$

Likewise, the error covariance prediction matrix P_k^- has the same form as in the equation except that F_k is replaced by F'_k and Q_k by

$$Q'_k \equiv DQ_k D^T = \text{diag}[Q_{pv}, Q_{hor}], \text{ where:}$$

$$Q_{pv} = \begin{bmatrix} S_p \frac{T^3}{3} & 0 & 0 & S_p \frac{T^2}{2} & 0 & 0 \\ 0 & S_p T & 0 & 0 & S_p \frac{T^2}{2} & 0 \\ S_p \frac{T^2}{2} & 0 & S_p T & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p \frac{T^3}{3} & 0 & S_p \frac{T^2}{2} \\ 0 & S_p \frac{T^2}{2} & 0 & 0 & S_p \frac{T^3}{3} & 0 \\ 0 & 0 & 0 & S_p \frac{T^2}{2} & 0 & S_p T \end{bmatrix},$$

S_p is the speed noise power spectrum,

$$Q_{hor} = \begin{bmatrix} Q_{hor,r} + Q_{hor,s1} & Q_{hor,r} & \dots & Q_{hor,r} \\ Q_{hor,r} & Q_{hor,r} + Q_{hor,s2} & \dots & Q_{hor,r} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{hor,r} & Q_{hor,r} & \dots & Q_{hor,r} + Q_{hor,sN} \end{bmatrix}.$$

The EK-SLAM estimator corrects the state errors by using the following EKF correction equations:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1}$$

$$\hat{x}_k = \hat{x}_k^- + K_k (Z_k - Z_k^-)$$

$$P_k = (I - K_k H_k) P_k^-$$

Where Z_k^- is the measurement estimated from the observation model. Z_k^- can also be estimated using the Jacobian H_k and the a priori state prediction $\hat{x}_k^-$: $Z_k^- = H_k \hat{x}_k^-$. The dimension of the covariance matrix of the measurement noise R_k depends on the number M of GNSS satellites processed and/or the number of 5G base stations the signals of which are received. $L = N + M$ is the total number of transmitters available. R_k is a matrix of dimension $2L \times 2L$ defined by:

$$R_k = \text{diag}[R_A^{GNSS}, R_B^{GNSS}, R_A^{5G}, R_B^{5G}]$$

Where:

15

$$R_A^{GNSS} = \begin{bmatrix} \sigma_{1DLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2DLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{MDLL}^2 \end{bmatrix},$$

$$R_B^{GNSS} = \begin{bmatrix} \sigma_{1FLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2FLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{MFLL}^2 \end{bmatrix},$$

$$R_A^{5G} = \begin{bmatrix} \sigma_{1DLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2DLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{NDLL}^2 \end{bmatrix},$$

$$R_B^{5G} = \begin{bmatrix} \sigma_{1FLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2FLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{NFLL}^2 \end{bmatrix},$$

$$\sigma_{DLL}^2 = \frac{B_n^{DLL} d}{2C/N_0}, \sigma_{FLL}^2 = \frac{B_n^{FLL} d}{\pi^2 T_n^2 C/N_0}, B_n^{DLL}:$$

DLL loop band of the GNSS or 5G signal followed, B_n^{FLL} :
FLL loop band of the GNSS or 5G signal followed, T_n :
coherent integration time of the GNSS or 5G DLL loop of
interest, C/N_0 : estimate of the carrier-to-noise ratio of the
GNSS or 5G channel of interest, and d: chip spacing
between the Early and Late correlators of the GNSS or 5G
DLL of interest.

In mapping mode, with $Z_k \equiv [z_{sv}^T, z_s^T]^T$, the corresponding
Jacobian matrix is obtained by:

$$H_k = [H_{sv}^T, H_s^T]^T$$

$$H_{sv} = \begin{bmatrix} \hat{a}_{sv1}^T & 0_{1 \times 3} & h_{hor}^T & 0_{1 \times 5N} \\ \vdots & \vdots & \vdots & \vdots \\ \hat{a}_{svM}^T & 0_{1 \times 3} & h_{hor}^T & 0_{1 \times 5N} \\ 0_{1 \times 3} & \hat{a}_{sv1}^T & h_{hor}^T & 0_{1 \times 5N} \\ \vdots & \vdots & \vdots & \vdots \\ 0_{1 \times 3} & \hat{a}_{svM}^T & h_{hor}^T & 0_{1 \times 5N} \end{bmatrix},$$

$$Z_k = \begin{bmatrix} Z_{s1} \\ \vdots \\ Z_{sN} \\ \hat{Z}_{s1} \\ \vdots \\ \hat{Z}_{sN} \end{bmatrix}$$

$$(z_{sn})_k = (\|r_r - r_{sn}\|^2 + c \cdot [\delta t_r - \delta t_{sn}])_k + (v_{sn})_k$$

$$(z_{sn})_k = \left(\frac{1 - \frac{1}{c} [V_{sn} - \dot{r}_r \cdot \hat{a}_{sn}^T]}{1 + (\delta t_r - \delta t_{sn})} \lambda \right)_k + (w_{sn})_k$$

$$\hat{X}_k'^- = F_k' \hat{X}_{k-1}'$$

$$\text{With } \hat{X}_k'^- = [\hat{r}_r'^T, \hat{r}_r'^T, \Delta \hat{x}_{hor1}'^T, \dots, \Delta \hat{x}_{horN}'^T]^T$$

16

-continued

$$H_s = \begin{bmatrix} \hat{a}_{s1}^T & 0_{1 \times 3} & h_{hor}^T & H_{s1} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \hat{a}_{sN}^T & 0_{1 \times 3} & h_{hor}^T & 0 & \dots & H_{sN} \\ 0_{1 \times 3} & \hat{a}_{s1}^T & h_{hor}^T & H_{s1}^a & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_{1 \times 3} & \hat{a}_{sN}^T & h_{hor}^T & 0 & \dots & H_{sN}^a \end{bmatrix}$$

$$\text{where } \hat{a}_{svm}^T \equiv \frac{\hat{r}_r - r_{svm}}{\|\hat{r}_r - r_{svm}\|},$$

$$h_{hor} \equiv [-1, 0]^T, h_{hor}^a \equiv [0, -1]^T, \hat{a}_{sn}^T \equiv \frac{\hat{r}_r - r_{sn}}{\|\hat{r}_r - r_{sn}\|},$$

$$H_{sn} \equiv [-\hat{a}_{sn}^T, -h_{hor}^T], \text{ and } H_{sn}^a \equiv [-\hat{a}_{sn}^T, -h_{hor}^a]^T.$$

The correction produces the a posteriori estimate $\hat{X}_k$ and the
a posteriori estimate of the corresponding error covariance
matrix P_k . The true states (positions, speeds, clocks) are
updated at the instant k by adding the a posteriori error
estimate $\hat{X}_k$ to the estimates of true states at the instant k-1
(i.e. $\hat{x}_k = \hat{x}_{k-1} + \hat{X}_k$). In 5G mode, only the 5G pseudo-distances
are available with $z = z_s$, and the Jacobian is given by:

$$H_k' = \begin{bmatrix} \hat{a}_{s1}^T & 0_{1 \times 3} & H_{s1}' & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \hat{a}_{sN}^T & 0_{1 \times 3} & 0 & \dots & H_{sN}' \\ 0_{1 \times 3} & \hat{a}_{s1}^T & H_{s1}'^a & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_{1 \times 3} & \hat{a}_{sN}^T & 0 & \dots & H_{sN}'^a \end{bmatrix}$$

$$\text{where } H_{sn}' \equiv [\hat{a}_{sn}^T, h_{hor}^T] \text{ and } H_{sn}'^a \equiv [-\hat{a}_{sn}^T, h_{hor}^a]^T.$$

The correction produces the a posteriori estimate $\hat{X}_k'$ and the
a posteriori estimate of the corresponding error covariance
matrix P_k' . The true states (positions, speeds, clocks) are
updated at the instant k by adding the a posteriori error
estimate $\hat{X}_k'$ to the estimates of true states at the instant k-1
(i.e. $\hat{x}_k = \hat{x}_{k-1} + \hat{X}_k'$).

In 5G mode, let N be the number of 5G base stations
received:

X represents 3+3+2*N states and Z represents N position
difference observations and N speed difference observations.

-continued

$$F'_k \equiv \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & O_{3 \times 2} & \dots & O_{3 \times 2} \\ O_{3 \times 3} & I_{3 \times 3} & O_{3 \times 2} & \dots & O_{3 \times 2} \\ O_{2 \times 3} & O_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} & \dots & O_{2 \times 3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ O_{2 \times 3} & O_{2 \times 3} & O_{2 \times 2} & \dots & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix}$$

 $Q =$

$$\begin{bmatrix} S_p \frac{T^3}{3} & 0 & 0 & S_p \frac{T^2}{2} & 0 & 0 \\ 0 & S_p T & 0 & 0 & S_p \frac{T^2}{2} & 0 \\ S_p \frac{T^2}{2} & 0 & S_p T & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p \frac{T^3}{3} & 0 & S_p \frac{T^2}{2} \\ 0 & S_p \frac{T^2}{2} & 0 & 0 & S_p \frac{T^3}{3} & 0 \\ 0 & 0 & 0 & S_p \frac{T^2}{2} & 0 & S_p T \end{bmatrix} \quad [0]$$

$$\begin{bmatrix} Q_{hor,s} + Q_{hor,s1} & Q_{hor,r} & \dots & Q_{hor,r} \\ Q_{hor,r} & Q_{hor,r} + Q_{hor,s2} & \dots & Q_{hor,r} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{hor,r} & Q_{hor,r} & Q_{hor,r} & Q_{hor,r} + Q_{hor,sN} \end{bmatrix}$$

$$Q_{hor,s} = \begin{bmatrix} S_{w\delta t_s} T + S_{w\delta t_s} \frac{T^3}{3} & S_{w\delta t_s} \frac{T^2}{2} \\ S_{w\delta t_s} \frac{T^2}{2} & S_{w\delta t_s} T \end{bmatrix}$$

$$Q_{hor,r} = \begin{bmatrix} S_{w\delta t_r} T + S_{w\delta t_s} \frac{T^3}{3} & S_{w\delta t_r} \frac{T^2}{2} \\ S_{w\delta t_r} \frac{T^2}{2} & S_{w\delta t_r} T \end{bmatrix}$$

The matrix H_k is of the form:

$$\begin{bmatrix} \hat{a}_{s1}^T & 0_{1 \times 3} & h_{hor}^T & H_{s1} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \hat{a}_{sN}^T & 0_{1 \times 3} & h_{hor}^T & 0 & \dots & H_{sN} \\ 0_{1 \times 3} & \hat{a}_{s1}^T & h_{hor}^T & h_{s1}^a & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_{1 \times 3} & \hat{a}_{sN}^T & h_{hor}^T & 0 & \dots & h_{sN}^a \end{bmatrix}$$

$$R = \begin{bmatrix} R_A^{sG} & [0] \\ [0] & R_A^{sG} \end{bmatrix}$$

$$R_A^{sG} = \begin{bmatrix} \sigma_{1DLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2DLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{NDLL}^2 \end{bmatrix}$$

$$R_B^{sG} = \begin{bmatrix} \sigma_{1FLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2FLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{NFLL}^2 \end{bmatrix}$$

In GNSS mode, standard mode, if M is the number of satellites the signals of which are received, X represents 3+3+2 states that are independent of the number of states and Z represents M position difference observations and M speed difference observations.

$$Z_k = \begin{bmatrix} Z_{sv1} \\ \vdots \\ Z_{svM} \\ \dot{Z}_{sv1} \\ \vdots \\ \dot{Z}_{svM} \end{bmatrix}$$

$$x_r = [r_r^T, \dot{r}_r^T, x_{hor,r}^T]^T$$

$$(x_r)_{k+1} = F_r(x_r)_k + (w_r)_k$$

$$(z_{svm})_k = (\|r_r - r_{svm}\|^2 + c \cdot [\delta t_r - \delta t_{svm}])_k + (v_{svm})_k$$

$$(z_{svm})_k = \left(\frac{1 - \frac{1}{c} [V_{svm} - \dot{r}_r \cdot \dot{a}_{svm}]}{1 + (\delta t_r - \delta t_{svm})} \lambda \right)_k + (w_{svm})_k$$

$$F_r = \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & 0_{3 \times 2} \\ 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix}$$

$$Q_r =$$

$$\begin{bmatrix} S_p T^3/3 & 0 & 0 & S_p T^2/2 & 0 & 0 & 0 & 0 \\ 0 & S_p T & 0 & 0 & S_p T^2/2 & 0 & 0 & 0 \\ S_p T^2/2 & 0 & S_p T & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p T^3/3 & 0 & S_p T^2/2 & 0 & 0 \\ 0 & S_p T^2/2 & 0 & 0 & S_p T^3/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p T^2/2 & 0 & S_p T & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & c^2 S_{w\delta t_r} T + c^2 S_{w\delta t_r} T^3/3 & c^2 S_{w\delta t_r} T^2/2 \\ 0 & 0 & 0 & 0 & 0 & 0 & c^2 S_{w\delta t_r} T^2/2 & c^2 S_{w\delta t_r} T \end{bmatrix}$$

The matrix H_k is of the form:

$$\begin{bmatrix} \hat{a}_{s1}^T & 0_{1 \times 3} & h_{hor}^T \\ \vdots & \vdots & \vdots \\ \hat{a}_{sM}^T & 0_{1 \times 3} & h_{hor}^T \\ 0_{1 \times 3} & \hat{a}_{s1}^T & h_{hor}^{aT} \\ \vdots & \vdots & \vdots \\ 0_{1 \times 3} & \hat{a}_{sM}^T & h_{hor}^{aT} \end{bmatrix}$$

$$\begin{bmatrix} R_A^{GNSS} & [0] \\ [0] & R_A^{GNSS} \end{bmatrix}$$

$$R_A^{GNSS} = \begin{bmatrix} \sigma_{1DLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2DLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{MDLL}^2 \end{bmatrix}$$

$$R_B^{GNSS} = \begin{bmatrix} \sigma_{1FLL}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2FLL}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{MFLL}^2 \end{bmatrix}$$

In mapping mode, X represents 3+3+2+(3+2)*N states and X is not the concatenation of the separate 5G and GNSS states. Z represents M observations of position difference and then M observations of speed difference+N observations of position difference and N of speed differences.

$$Z_k = \begin{bmatrix} Z_{sv1} \\ \vdots \\ Z_{svM} \\ \dot{Z}_{sv1} \\ \vdots \\ \dot{Z}_{svM} \\ Z_{s1} \\ \vdots \\ Z_{sN} \\ \dot{Z}_{s1} \\ \vdots \\ \dot{Z}_{sN} \end{bmatrix}$$

$$\hat{X}_k = [\hat{p}_r^T, \hat{\dot{p}}_r^T, \hat{x}_{hor,r}^T, \hat{p}_{s1}^T, \hat{x}_{hor,s1}^T, \dots, \hat{p}_{sN}^T, \hat{x}_{hor,sN}^T]^T$$

$$F_k = \begin{bmatrix} I_{3 \times 3} & T \cdot I_{3 \times 3} & 0_{3 \times 2} & 0_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 3} & 0_{3 \times 2} \\ 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 2} & 0_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} & 0_{2 \times 3} & 0_{2 \times 2} & \dots & 0_{2 \times 3} & 0_{2 \times 2} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 2} & I_{3 \times 3} & 0_{3 \times 2} & \dots & 0_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & 0_{2 \times 2} & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} & \dots & 0_{2 \times 3} & 0_{2 \times 2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 2} & 0_{3 \times 3} & 0_{3 \times 2} & \dots & I_{3 \times 3} & 0_{3 \times 2} \\ 0_{2 \times 3} & 0_{2 \times 3} & 0_{2 \times 2} & 0_{2 \times 3} & 0_{2 \times 2} & \dots & 0_{2 \times 3} & \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{bmatrix}$$

$$Q = \begin{bmatrix} [Q_r] & [0] & [0] & [0] & [0] \\ [0] & [Q_{s1}] & [0] & [0] & [0] \\ [0] & [0] & \ddots & [0] & [0] \\ [0] & [0] & [0] & \ddots & [0] \\ [0] & [0] & [0] & [0] & [Q_{sN}] \end{bmatrix}$$

$$Q_r =$$

$$\begin{bmatrix} S_p T^3/3 & 0 & 0 & S_p T^2/2 & 0 & 0 & 0 & 0 \\ 0 & S_p T & 0 & 0 & S_p T^2/2 & 0 & 0 & 0 \\ S_p T^2/2 & 0 & S_p T & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p T^3/3 & 0 & S_p T^2/2 & 0 & 0 \\ 0 & S_p T^2/2 & 0 & 0 & S_p T^3/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_p T^2/2 & 0 & S_p T & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & c^2 S_{w\dot{\theta}_{tr}} T + c^2 S_{w\ddot{\theta}_{tr}} T^3/3 & c^2 S_{w\ddot{\theta}_{tr}} T^2/2 \\ 0 & 0 & 0 & 0 & 0 & 0 & c^2 S_{w\dot{\theta}_{tr}} T^2/2 & c^2 S_{w\ddot{\theta}_{tr}} T \end{bmatrix}$$

$$Q_{si} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & C^2 & C^2 \cdot T \\ 0 & 0 & 0 & 1 & C^2 \end{bmatrix}$$

$$H_k = \begin{bmatrix} H_{sv} \\ H_s \end{bmatrix}$$

23

The matrix H_k is of the form:

$$H_{sv} = \begin{bmatrix} \hat{a}_{sv1}^T & 0_{1 \times 3} & h_{hor}^T & 0_{1 \times 5N} \\ \vdots & \vdots & \vdots & \vdots \\ \hat{a}_{svM}^T & 0_{1 \times 3} & h_{hor}^T & 0_{1 \times 5N} \\ 0_{1 \times 3} & \hat{a}_{sv1}^T & h_{hor}^T & 0_{1 \times 5N} \\ \vdots & \vdots & \vdots & \vdots \\ 0_{1 \times 3} & \hat{a}_{svM}^T & h_{hor}^T & 0_{1 \times 5N} \end{bmatrix}$$

$$H_s = \begin{bmatrix} \hat{a}_{s1}^T & 0_{1 \times 3} & h_{hor}^T & H_{s1} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \hat{a}_{sN}^T & 0_{1 \times 3} & h_{hor}^T & 0 & \dots & H_{sN} \\ 0_{1 \times 3} & \hat{a}_{s1}^T & h_{hor}^T & h_{s1}^a & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_{1 \times 3} & \hat{a}_{sN}^T & h_{hor}^T & 0 & \dots & h_{sN}^a \end{bmatrix}$$

Where $\hat{a}_{svm}^T \cong \frac{\hat{r}_r - r_{svm}}{\|\hat{r}_r - r_{svm}\|}$, $h_{hor} \cong [-1, 0]^T$, $h_{hor}^a \cong [0, -1]^T$

$$\hat{a}_{sn}^T \cong \frac{\hat{r}_r - r_{sn}}{\|\hat{r}_r - r_{sn}\|}, H_{sn} \cong [-\hat{a}_{sn}^T, -h_{hor}^T], et$$

$$H_{sn}^a \cong [-\hat{a}_{sn}^T, -h_{hor}^T]$$

$$R = \begin{bmatrix} R_A^{GNSS} & [0] & [0] & [0] \\ [0] & R_B^{GNSS} & [0] & [0] \\ [0] & [0] & R_A^{SG} & [0] \\ [0] & [0] & [0] & R_B^{SG} \end{bmatrix}$$

FIG. 12 illustrates schematically a method implemented by the processing module for selecting the operating mode of a navigator.

The step E120 corresponds to an operating mode wherein only GNSS signals are measured, the operating mode of the navigator is the standard GNSS mode. This mode corresponds for example to the case where the carrier is on the open sea.

At the step E121, the processing module checks whether the signals transmitted by the 5G base stations are being measured.

If so, the processing module passes to the step E122. If not, the processing module returns to the step E120.

The step E122 corresponds to an operating mode wherein GNSS and 5G signals are measured, the operating mode of the navigator is the mapping mode. This mode corresponds for example to the case where the carrier is approaching a port.

At the step E123, the processing module checks whether the GNSS signals cannot be used.

If so, the processing module passes to the step E124, if not, the processing module returns to the step E122.

The step E124 corresponds to a 5G operating mode. This mode corresponds for example to the case where the carrier is docked.

At the step E125, the processing module checks whether the 5G signals cannot be used.

If so, the processing module returns to the step E120. If not, the processing module returns to the step E124.

FIG. 13 illustrates schematically a method implemented by a navigator of the loose coupling system between an INS system and a GNSS/5G tight coupling system.

At the step E130, the initial estimate of the state vector $\hat{X}_0^-$ and the initial estimate of the error covariance are obtained.

24

At the following step E131, the operating mode of the navigator is obtained as well as the measurements Z_k , Z'_k or Z''_k .

At the step E132, the Jacobian matrices and the process noise covariance matrices are obtained.

If the operating mode is the mapping mode, the matrices F_k , H_k , R_k and Q_k are obtained.

If the operating mode is the 5G mode, the matrices F'_k , H'_k , R'_k and Q'_k are obtained.

If the operating mode is the standard GNSS mode, the matrices F''_k , H''_k , R''_k and Q''_k are obtained.

At the steps E133 to E135, a correction is made using the matrices obtained.

More precisely, a calculation of the Kalman gain is made at the step E133, a calculation of the correction of the a priori estimate of the new measurement is made at the step E134 and a calculation the error covariance matrix of the estimate is calculated.

Thus, when the mode is the mapping mode, the following matrices are calculated:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1}$$

$$\hat{X}_k = \hat{X}_k^- + K_k (Z_k - Z_k^-)$$

$$P_k = (I - K_k H_k) P_k^-$$

When the mode is the 5G mode, the following matrices are calculated:

$$K'_k = P'_k^- H'^T_k (H'_k P'_k^- H'^T_k + R'_k)^{-1}$$

$$\hat{X}'_k = \hat{X}'_k^- + K'_k (Z'_k - Z'^-_{k-1})$$

$$P'_k = (I - K'_k H'_k) P'_k^-$$

When the mode is the standard GNSS mode, the following matrices are calculated:

$$K''_k = P''_k^- H''^T_k (H''_k P''_k^- H''^T_k + R''_k)^{-1}$$

$$\hat{X}''_k = \hat{X}''_k^- + K''_k (Z''_k - Z''^-_{k-1})$$

$$P''_k = (I - K''_k H''_k) P''_k^-$$

At the end of the correction, the prediction is made at the steps E136 to E138. At the step E136, a calculation of the prediction of the state vector is made. At the step E137, the prediction of the measurement vector is made and at the step E138 the prediction of the error covariance matrix is calculated.

Thus, when the mode is the mapping mode, the following predictions are calculated:

$$\hat{X}_{k+1}^- = F_k \hat{X}_k$$

$$Z_{k+1}^- = F_k H_k \hat{X}_{k+1}^-$$

$$P_{k+1}^- = F_k P_k F_k^T + Q_k$$

When the mode is the 5G mode, the following predictions are calculated:

$$\hat{X}'_{k+1}^- = F'_k \hat{X}'_k$$

$$Z'_{k+1}^- = F'_k H'_k \hat{X}'_{k+1}^-$$

$$P'_{k+1}^- = F'_k P'_k F'^T_k + Q'_k$$

When the mode is the standard GNSS mode, the following predictions are calculated:

$$\hat{X}''_{k+1}^- = F''_k \hat{X}''_k$$

$$Z''_{k+1}^- = F''_k H''_k \hat{X}''_{k+1}^-$$

$$P''_{k+1}^- = F''_k P''_k F''^T_k + Q''_k$$

25

The invention claimed is:

1. A navigation device comprising:

an inertial navigation system coupled with a satellite navigation system wherein information supplied by the satellite navigation system is used for adjusting the inertial navigation system; and

circuitry causing the device to perform measuring signals coming from base stations of a wireless cellular network,

wherein the satellite navigation system and the circuitry are coupled with each other by a tight coupling to form a satellite navigator and/or base-station navigator implementing an estimator with simultaneous localisation and mapping,

wherein the satellite navigation system and the circuitry are coupled to the inertial navigation system by a loose coupling for adjusting the inertial navigation system, and

wherein the satellite and/or base station navigator operates in three operating modes, a first mode being a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network and signals from the satellites are available, a second mode being a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network are available, and a third mode being a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals of the signals of the satellites are available.

2. The device according to claim 1, wherein the device further comprises a matrix-generating module that forms a plurality of matrices comprising a transition matrix, a Jacobian matrix, a measurement-noise covariance matrix and a process-noise covariance matrix.

3. The device according to claim 2, wherein, for each operating mode, the matrix-generating module forms a transition matrix, a Jacobian matrix, a measurement-noise covariance matrix and a process-noise covariance matrix that are different from the matrices formed for the other modes.

26

4. The device according to claim 1, wherein the navigation device comprises a Kalman filter.

5. A navigation method comprising an inertial navigation system coupled with a satellite navigation system, the information supplied by the satellite navigation system being used for adjusting the inertial navigation system, wherein said method comprises:

measuring signals coming from base stations of a wireless cellular network;

coupling by tight coupling of the satellite navigation system with means for measuring signals coming from base stations of the wireless cellular network to form a satellite navigator and/or base-station navigator implementing an estimator with simultaneous localisation and mapping;

coupling by loose coupling of the satellite navigation system and of the means for measuring signals coming from the base station of the wireless cellular network to the inertial navigation system to adjust the inertial navigation system; and

determining an operating mode of the satellite and/or base-station navigator among three operating modes such that a first mode is a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network and signals from the satellites are available, a second mode is a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals coming from base stations of the wireless cellular network are available and a third mode is a mode wherein pseudo-distances, pseudo-speeds and carrier-to-noise ratios of the signals of the signals of the satellites are available.

6. The method according to claim 5, wherein said method causes the device to perform generating a matrix that forms a plurality of matrices comprising a transition matrix, a Jacobian matrix, a measurement-noise covariance matrix and a process-noise covariance matrix.

* * * * *